

COMPLETE JAVA PROGRAMMING MASTERCLASS

30 CORE TOPICS WITH DETAILED THEORY, CODE EXAMPLES, 750
PRACTICE QUESTIONS & ANSWER KEYS

Language: English

Target: Beginner to Advanced Developers

Format: Comprehensive Reference & Practice Guide

Topic 1: 1. Introduction to Java & JVM Architecture

1. Theory

Java is a high-level, object-oriented, robust, secure, and platform-independent programming language developed by Sun Microsystems (now owned by Oracle) in 1995. The core philosophy of Java is 'Write Once, Run Anywhere' (WORA), which is achieved through the Java Virtual Machine (JVM). When a Java source file (.java) is compiled, it is converted into bytecode (.class) by the Java Compiler (javac). This bytecode is platform-independent and can be executed on any operating system that has a compatible JVM installed. The Java Development Kit (JDK) provides the tools required to write and compile Java programs, while the Java Runtime Environment (JRE) provides the libraries and JVM needed to run them.

2. Code Example

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Welcome to Java Programming Masterclass!");  
        System.out.println("Java Version: " + System.getProperty("java.version"));  
    }  
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 1. Introduction to Java & JVM Architecture?

- A) Option A for 1. Introduction to Java & JVM Architecture statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 2: 2. Variables, Data Types & Type Casting

1. Theory

Variables are containers for storing data values in Java. Java is a statically typed language, meaning all variables must be declared before they can be used. Data types are divided into two categories: Primitive data types (such as int, double, char, boolean, byte, short, long, float) which store raw values directly in memory, and Non-primitive/Reference data types (such as Strings, Arrays, and Classes) which store references to objects. Type casting is the process of converting one data type into another. Widening casting (implicit) converts a smaller type to a larger type size automatically (e.g., int to double), whereas Narrowing casting (explicit) must be done manually by placing the target type in parentheses (e.g., double to int).

2. Code Example

```
public class DataTypesDemo {
    public static void main(String[] args) {
        int originalInt = 100;
        double convertedDouble = originalInt; // Widening casting

        double pi = 3.14159;
        int truncatedPi = (int) pi; // Narrowing casting

        System.out.println("Converted Double: " + convertedDouble);
        System.out.println("Truncated Int: " + truncatedPi);
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

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- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 2. Variables, Data Types & Type Casting?

- A) Option A for 2. Variables, Data Types & Type Casting statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 3: 3. Operators in Java

1. Theory

Operators are symbols used to perform operations on variables and values. Java provides several categories of operators: Arithmetic operators (+, -, *, /, %), Assignment operators (=, +=, -=, *=, /=), Relational operators (==, !=, >, <, >=, <=), Logical operators (&&, ||, !), Bitwise operators (&, |, ^, ~, <<, >>), and the Ternary operator (condition ? true_block : false_block). Operator precedence determines the order in which operators are evaluated in an expression, where multiplication and division take precedence over addition and subtraction.

2. Code Example

```
public class OperatorDemo {
    public static void main(String[] args) {
        int a = 10, b = 20;
        boolean condition = (a < b) && (b > 15);

        int max = (a > b) ? a : b; // Ternary operator

        System.out.println("Condition Result: " + condition);
        System.out.println("Maximum Value: " + max);
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
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- A) Option A for 3. Operators in Java statement
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- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

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- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 3. Operators in Java?

- A) Option A for 3. Operators in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 4: 4. Control Statements & Decision Making

1. Theory

Control flow statements dictate the order in which statements are executed in a Java program. Conditional statements allow execution paths to branch based on boolean expressions. The primary conditional constructs include 'if', 'if-else', 'nested if', 'if-else-if' ladders, and 'switch' selection blocks. The switch statement supports byte, short, int, char, String, and enum types, evaluating a target expression against multiple case labels and supporting optional break statements to terminate execution flow.

2. Code Example

```
public class ControlDemo {  
    public static void main(String[] args) {  
        int score = 85;  
        if (score >= 90) {  
            System.out.println("Grade: A");  
        } else if (score >= 75) {  
            System.out.println("Grade: B");  
        } else {  
            System.out.println("Grade: C");  
        }  
    }  
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 4. Control Statements & Decision Making?

- A) Option A for 4. Control Statements & Decision Making statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 5: 5. Looping & Iteration Constructs

1. Theory

Iteration statements allow code blocks to be executed repeatedly based on a loop termination condition. Java supports entry-controlled loops ('for' and 'while'), an exit-controlled loop ('do-while'), and an enhanced 'for-each' loop designed for iterating through collections and arrays. Loop control statements such as 'break' (to exit the loop entirely) and 'continue' (to skip the current iteration) provide fine-grained control over execution flow.

2. Code Example

```
public class LoopDemo {
    public static void main(String[] args) {
        // For loop example
        for (int i = 1; i <= 3; i++) {
            System.out.println("Iteration: " + i);
        }

        int count = 2;
        do {
            System.out.println("Do-While Count: " + count);
            count--;
        } while (count > 0);
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 5. Looping & Iteration Constructs?

- A) Option A for 5. Looping & Iteration Constructs statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 6: 6. Object-Oriented Programming (Classes & Objects)

1. Theory

Object-Oriented Programming (OOP) is a programming paradigm based on the concept of 'objects', which contain data (fields/attributes) and code (methods/behaviors). A class serves as a blueprint or template from which individual objects are instantiated using the 'new' keyword. Java classes encapsulate state and behavior, supporting constructors for object initialization, 'this' references to current object instances, and member access controls.

2. Code Example

```
class Car {
    String brand;
    int speed;

    // Constructor
    public Car(String b, int s) {
        brand = b;
        speed = s;
    }

    public void displayInfo() {
        System.out.println("Brand: " + brand + ", Speed: " + speed + " km/h");
    }
}

public class OOPDemo {
    public static void main(String[] args) {
        Car myCar = new Car("Tesla", 120);
        myCar.displayInfo();
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
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- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
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- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
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- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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Q18. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
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- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
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- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
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- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

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- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 6. Object-Oriented Programming (Classes & Objects)?

- A) Option A for 6. Object-Oriented Programming (Classes & Objects) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 7: 7. Four Pillars of OOP: Inheritance & Polymorphism

1. Theory

Inheritance allows a class (subclass/derived class) to inherit fields and methods from another class (superclass/base class) using the 'extends' keyword, promoting code reusability. Polymorphism ('many forms') enables objects to be treated as instances of their parent class rather than their actual class. Compile-time polymorphism is achieved through method overloading, while runtime polymorphism is achieved through method overriding, where a subclass provides a specific implementation of a method defined in its superclass.

2. Code Example

```
class Animal {
    public void sound() {
        System.out.println("Animal makes a sound");
    }
}

class Dog extends Animal {
    @Override
    public void sound() {
        System.out.println("Dog barks");
    }
}

public class PolymorphismDemo {
    public static void main(String[] args) {
        Animal myDog = new Dog(); // Runtime Polymorphism
        myDog.sound();
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 7. Four Pillars of OOP: Inheritance & Polymorphism?

- A) Option A for 7. Four Pillars of OOP: Inheritance & Polymorphism statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 8: 8. Encapsulation & Abstraction

1. Theory

Encapsulation is the mechanism of wrapping data (variables) and code acting on the data (methods) together as a single unit, achieved by declaring class variables as private and providing public getter and setter methods. Abstraction is the concept of hiding implementation details and showing only essential features to the user, implemented in Java using abstract classes (which can have both abstract and concrete methods) and interfaces (which define contracts with fully abstract methods or default methods).

2. Code Example

```
interface Vehicle {
    void start();
}

class Bike implements Vehicle {
    public void start() {
        System.out.println("Bike engine started with kick.");
    }
}

public class AbstractionDemo {
    public static void main(String[] args) {
        Vehicle v = new Bike();
        v.start();
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

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- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

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- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

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- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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Q16. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 8. Encapsulation & Abstraction?

- A) Option A for 8. Encapsulation & Abstraction statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 9: 9. Arrays & Multidimensional Arrays

1. Theory

An array is a data structure used to store a fixed-size collection of elements of the same data type sequentially in memory. Array indices start at 0 and go up to length-1. Java also supports multidimensional arrays (arrays of arrays), commonly used for matrices and grid structures. Arrays have a built-in 'length' property to retrieve their total capacity.

2. Code Example

```
public class ArrayDemo {
    public static void main(String[] args) {
        int[] scores = {90, 85, 88, 92, 95};
        int sum = 0;

        for (int score : scores) {
            sum += score;
        }

        double average = (double) sum / scores.length;
        System.out.println("Average Score: " + average);
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
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- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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Q18. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

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- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 9. Arrays & Multidimensional Arrays?

- A) Option A for 9. Arrays & Multidimensional Arrays statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 10: 10. Strings & String Manipulation

1. Theory

Strings in Java are objects represented by the 'java.lang.String' class. Unlike many other languages, Java Strings are immutable, meaning once created, their values cannot be changed; any modification creates a new String object in the String Constant Pool. For mutable string operations, Java provides 'StringBuilder' and 'StringBuffer' classes, which are much more efficient for frequent concatenation and string modifications in loops.

2. Code Example

```
public class StringDemo {
    public static void main(String[] args) {
        String s1 = "Hello";
        String s2 = "World";
        String result = s1 + " " + s2;

        StringBuilder sb = new StringBuilder("Java");
        sb.append(" Programming");

        System.out.println("Result: " + result);
        System.out.println("StringBuilder: " + sb.toString());
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 10. Strings & String Manipulation?

- A) Option A for 10. Strings & String Manipulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 11: 11. Exception Handling & Robustness

1. Theory

Exception handling is a powerful mechanism in Java to handle runtime errors and maintain normal application flow. Exceptions are divided into checked exceptions (verified at compile-time, e.g., `IOException`) and unchecked exceptions (occur at runtime, e.g., `NullPointerException`, `ArithmeticException`). Java uses 'try', 'catch', 'finally', 'throw', and 'throws' blocks to intercept, manage, and propagate errors gracefully.

2. Code Example

```
public class ExceptionDemo {
    public static void main(String[] args) {
        try {
            int data = 50 / 0;
        } catch (ArithmeticException e) {
            System.out.println("Caught Exception: Division by zero is not allowed.");
        } finally {
            System.out.println("Finally block always executes.");
        }
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 11. Exception Handling & Robustness?

- A) Option A for 11. Exception Handling & Robustness statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 12: 12. Multithreading & Concurrency

1. Theory

Multithreading is the ability of a CPU to execute multiple threads of execution concurrently, maximizing resource utilization. In Java, threads can be created either by extending the 'Thread' class or by implementing the 'Runnable' interface. The life cycle of a thread includes New, Runnable, Blocked/Waiting, Timed Waiting, and Terminated states. Synchronization mechanisms prevent thread interference and memory consistency errors in multi-threaded environments.

2. Code Example

```
class SampleThread extends Thread {
    public void run() {
        System.out.println("Thread running concurrently: " +
Thread.currentThread().getId());
    }
}

public class ThreadDemo {
    public static void main(String[] args) {
        SampleThread t1 = new SampleThread();
        t1.start();
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 12. Multithreading & Concurrency?

- A) Option A for 12. Multithreading & Concurrency statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 13: 13. Java Collections Framework (Lists & Sets)

1. Theory

The Java Collections Framework provides a unified architecture for storing and manipulating groups of objects. The 'List' interface (implemented by ArrayList and LinkedList) maintains ordered collections allowing duplicate elements. The 'Set' interface (implemented by HashSet and TreeSet) models mathematical sets containing unique elements only. Generics ensure type safety across collection elements without requiring explicit casting.

2. Code Example

```
import java.util.ArrayList;
import java.util.List;

public class CollectionDemo {
    public static void main(String[] args) {
        List languages = new ArrayList<>();
        languages.add("Java");
        languages.add("Python");
        languages.add("C++");

        for (String lang : languages) {
            System.out.println("Language: " + lang);
        }
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 13. Java Collections Framework (Lists & Sets)?

- A) Option A for 13. Java Collections Framework (Lists & Sets) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 14: 14. Java Collections Framework (Maps & Iterators)

1. Theory

Maps store data in key-value pairs where each key is unique. The 'Map' interface is implemented by classes such as HashMap, LinkedHashMap, and TreeMap. Iterators provide a standard mechanism to traverse through collection elements sequentially, supporting safe element removal during iteration. Sorting mechanisms are facilitated by the 'Comparable' and 'Comparator' interfaces.

2. Code Example

```
import java.util.HashMap;
import java.util.Map;

public class MapDemo {
    public static void main(String[] args) {
        Map studentScores = new HashMap<>();
        studentScores.put("Alice", 92);
        studentScores.put("Bob", 85);

        for (Map.Entry entry : studentScores.entrySet()) {
            System.out.println(entry.getKey() + ": " + entry.getValue());
        }
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 14. Java Collections Framework (Maps & Iterators)?

- A) Option A for 14. Java Collections Framework (Maps & Iterators) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 15: 15. Java I/O & File Handling

1. Theory

Java I/O packages provide classes to read and write data streams, files, and console inputs. Character streams ('FileReader', 'FileWriter', 'BufferedReader') are designed for handling character data, while byte streams ('FileInputStream', 'FileOutputStream') manage raw binary data. Modern Java NIO (New I/O) provides high-performance file operations, asynchronous channels, and advanced path management.

2. Code Example

```
import java.io.File;
import java.io.IOException;

public class FileDemo {
    public static void main(String[] args) {
        try {
            File myObj = new File("testfile.txt");
            if (myObj.createNewFile()) {
                System.out.println("File created: " + myObj.getName());
            } else {
                System.out.println("File already exists.");
            }
        } catch (IOException e) {
            System.out.println("An error occurred.");
            e.printStackTrace();
        }
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 15. Java I/O & File Handling?

- A) Option A for 15. Java I/O & File Handling statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 16: 16. Java Generics & Type Safety

1. Theory

Generics enable types (classes and methods) to be parameters when defining classes, interfaces, and methods. This eliminates the need for type casting and enables detection of compile-time bugs. Generic methods and wildcard types ("?", "*", ") provide extreme flexibility in API design while maintaining strict type safety.

2. Code Example

```
class Box {
    private T t;
    public void set(T t) { this.t = t; }
    public T get() { return t; }
}

public class GenericsDemo {
    public static void main(String[] args) {
        Box stringBox = new Box<>();
        stringBox.set("Generic Java Programming");
        System.out.println(stringBox.get());
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

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- A) Option A for 16. Java Generics & Type Safety statement
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- C) Option C for alternative structure
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- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 16. Java Generics & Type Safety?

- A) Option A for 16. Java Generics & Type Safety statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 17: 17. Lambda Expressions & Functional Interfaces

1. Theory

Introduced in Java 8, lambda expressions bring functional programming capabilities to Java, enabling developers to write concise code blocks that can be treated as method arguments. A functional interface is an interface containing exactly one abstract method (annotated with '@FunctionalInterface'), serving as the target type for lambda expressions and method references.

2. Code Example

```
@FunctionalInterface
interface Calculator {
    int operate(int a, int b);
}

public class LambdaDemo {
    public static void main(String[] args) {
        Calculator add = (a, b) -> a + b;
        System.out.println("Addition Result: " + add.operate(10, 20));
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 17. Lambda Expressions & Functional Interfaces?

- A) Option A for 17. Lambda Expressions & Functional Interfaces statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 18: 18. Stream API & Data Processing

1. Theory

The Stream API facilitates declarative processing of data collections. Streams support intermediate operations (such as filter, map, sorted) which return another stream, and terminal operations (such as collect, forEach, reduce, count) which produce a result or side-effect. Stream pipelines enable clean, parallelizable, and efficient data filtering and transformation.

2. Code Example

```
import java.util.Arrays;
import java.util.List;

public class StreamDemo {
    public static void main(String[] args) {
        List numbers = Arrays.asList(1, 2, 3, 4, 5, 6);
        int sumOfSquares = numbers.stream()
                                .filter(n -> n % 2 == 0)
                                .mapToInt(n -> n * n)
                                .sum();
        System.out.println("Sum of squared evens: " + sumOfSquares);
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 18. Stream API & Data Processing?

- A) Option A for 18. Stream API & Data Processing statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 19: 19. Java Reflection API & Annotations

1. Theory

Reflection is a feature in Java that allows an executing program to examine or 'inspect' itself and manipulate internal properties of its classes, methods, fields, and constructors at runtime. Annotations provide metadata about program elements without directly affecting their execution semantics, widely used in modern frameworks like Spring and Hibernate for configuration and dependency injection.

2. Code Example

```
import java.lang.reflect.Method;

public class ReflectionDemo {
    public static void main(String[] args) {
        try {
            Class c = Class.forName("java.lang.String");
            Method[] methods = c.getDeclaredMethods();
            System.out.println("Total methods in String class: " + methods.length);
        } catch (ClassNotFoundException e) {
            e.printStackTrace();
        }
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 19. Java Reflection API & Annotations?

- A) Option A for 19. Java Reflection API & Annotations statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 20: 20. JDBC & Database Connectivity

1. Theory

Java Database Connectivity (JDBC) is an API that defines how a client may access databases in Java. It provides methods to query and update data in relational databases. JDBC driver management, connection establishment via 'DriverManager', SQL statement execution ('Statement', 'PreparedStatement'), and result set processing form the bedrock of database interaction in enterprise Java applications.

2. Code Example

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class JDBCdemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/testdb";
        try (Connection conn = DriverManager.getConnection(url, "root", "password")) {
            if (conn != null) {
                System.out.println("Connected to the database successfully!");
            }
        } catch (SQLException e) {
            System.out.println("Database connection failed: " + e.getMessage());
        }
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 20. JDBC & Database Connectivity?

- A) Option A for 20. JDBC & Database Connectivity statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 21: 21. Object Class Methods & Cloning

1. Theory

Every class in Java implicitly inherits from 'java.lang.Object', making its methods—such as 'toString()', 'equals()', 'hashCode()', 'getClass()', and 'finalize()'—globally available. Object cloning via the 'Cloneable' interface and 'clone()' method allows creating exact copies of objects, distinguishing between shallow and deep copies.

2. Code Example

```
class Student implements Cloneable {
    String name;
    public Student(String name) { this.name = name; }

    public Object clone() throws CloneNotSupportedException {
        return super.clone();
    }
}

public class CloneDemo {
    public static void main(String[] args) {
        try {
            Student s1 = new Student("Alice");
            Student s2 = (Student) s1.clone();
            System.out.println("Cloned Student Name: " + s2.name);
        } catch (CloneNotSupportedException e) {
            e.printStackTrace();
        }
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 21. Object Class Methods & Cloning?

- A) Option A for 21. Object Class Methods & Cloning statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 22: 22. Java Garbage Collection & Memory Management

1. Theory

Java manages memory automatically through Garbage Collection (GC). The JVM heap is divided into generations: Young Generation (Eden and Survivor spaces) and Old/Tenured Generation. Objects that are no longer reachable by any active reference in the program become eligible for garbage collection, freeing up heap memory and preventing memory leaks without manual 'free()' calls.

2. Code Example

```
public class MemoryDemo {
    public static void main(String[] args) {
        Runtime runtime = Runtime.getRuntime();
        long memoryBefore = runtime.freeMemory();

        // Creating objects to consume memory
        String[] arr = new String[100000];

        long memoryAfter = runtime.freeMemory();
        System.out.println("Memory consumed by array: " + (memoryBefore - memoryAfter) + "
bytes");
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 22. Java Garbage Collection & Memory Management?

- A) Option A for 22. Java Garbage Collection & Memory Management statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 23: 23. Nested & Inner Classes

1. Theory

Java allows defining a class within another class, known as nested classes. They are categorized into static nested classes and non-static inner classes (including member inner classes, local inner classes inside methods, and anonymous inner classes). Inner classes have special access privileges to the enclosing class's private members, enhancing encapsulation and event-driven architecture design.

2. Code Example

```
class OuterClass {
    private int outerField = 10;

    class InnerClass {
        void display() {
            System.out.println("Outer field accessed from inner class: " + outerField);
        }
    }
}

public class InnerClassDemo {
    public static void main(String[] args) {
        OuterClass outer = new OuterClass();
        OuterClass.InnerClass inner = outer.new InnerClass();
        inner.display();
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 23. Nested & Inner Classes?

- A) Option A for 23. Nested & Inner Classes statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 24: 24. Enums & Enumerated Types

1. Theory

An enum in Java is a special data type that enables for a variable to be a set of predefined constants. Enums improve type safety, represent fixed categories cleanly, and can contain fields, constructors, and methods. They implicitly inherit from 'java.lang.Enum' and support switch-case matching seamlessly.

2. Code Example

```
enum Level {
    LOW, MEDIUM, HIGH
}

public class EnumDemo {
    public static void main(String[] args) {
        Level myVar = Level.MEDIUM;
        switch(myVar) {
            case LOW: System.out.println("Low level"); break;
            case MEDIUM: System.out.println("Medium level"); break;
            case HIGH: System.out.println("High level"); break;
        }
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 24. Enums & Enumerated Types?

- A) Option A for 24. Enums & Enumerated Types statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 25: 25. Packages, Access Modifiers & Encapsulation

1. Theory

Packages are namespaces used to organize related classes and interfaces in Java, preventing naming conflicts and facilitating access protection. Java provides four access modifiers: 'private' (accessible only within the class), 'default' (within package), 'protected' (within package and subclasses), and 'public' (globally accessible across applications).

2. Code Example

```
package com.masterclass.java;

public class AccessModifierDemo {
    public int publicVar = 1;
    private int privateVar = 2;

    public void showPrivate() {
        System.out.println("Private Var: " + privateVar);
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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Q14. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

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- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 25. Packages, Access Modifiers & Encapsulation?

- A) Option A for 25. Packages, Access Modifiers & Encapsulation statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 26: 26. Regular Expressions (Regex) in Java

1. Theory

Regular expressions provide a powerful mechanism for pattern matching, text searching, and string validation in Java. The 'java.util.regex' package contains the 'Pattern' and 'Matcher' classes, which allow compiling regular expression patterns and applying them against target strings for validation (e.g., email or phone number checking).

2. Code Example

```
import java.util.regex.Matcher;
import java.util.regex.Pattern;

public class RegexDemo {
    public static void main(String[] args) {
        Pattern pattern = Pattern.compile("[A-Za-z0-9+_.-]+@(.+)$");
        Matcher matcher = pattern.matcher("test@example.com");
        System.out.println("Email Match Result: " + matcher.matches());
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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Q9. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
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- A) Option A for 26. Regular Expressions (Regex) in Java statement
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Q16. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

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- C) Option C for alternative structure
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- A) Option A for 26. Regular Expressions (Regex) in Java statement
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Q18. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
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- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 26. Regular Expressions (Regex) in Java?

- A) Option A for 26. Regular Expressions (Regex) in Java statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 27: 27. Java Time & Date API (java.time)

1. Theory

Introduced in Java 8, the 'java.time' package provides a comprehensive, thread-safe, and immutable date and time API to replace legacy classes like 'java.util.Date' and 'java.util.Calendar'. Key classes include 'LocalDate', 'LocalTime', 'LocalDateTime', 'ZonedDateTime', and 'Duration' for precise time arithmetic.

2. Code Example

```
import java.time.LocalDate;
import java.time.LocalTime;

public class TimeDemo {
    public static void main(String[] args) {
        LocalDate date = LocalDate.now();
        LocalTime time = LocalTime.now();
        System.out.println("Current Date: " + date);
        System.out.println("Current Time: " + time);
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 27. Java Time & Date API (java.time)?

- A) Option A for 27. Java Time & Date API (java.time) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 28: 28. Optional Class & Avoiding NullPointerExceptions

1. Theory

The 'java.util.Optional' class is a container object used to represent the presence or absence of a non-null value. Introduced to mitigate the notorious NullPointerException (dubbed 'The Billion Dollar Mistake'), Optional provides functional methods like 'orElse()', 'map()', and 'ifPresent()' to handle null references gracefully.

2. Code Example

```
import java.util.Optional;

public class OptionalDemo {
    public static void main(String[] args) {
        String[] words = new String[10];
        Optional checkNull = Optional.ofNullable(words[5]);

        if (checkNull.isPresent()) {
            System.out.println(checkNull.get());
        } else {
            System.out.println("Value is not available (null safely handled).");
        }
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 28. Optional Class & Avoiding NullPointerExceptions?

- A) Option A for 28. Optional Class & Avoiding NullPointerExceptions statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 29: 29. Serialization & Object Persistence

1. Theory

Serialization is the process of converting an object's state into a byte stream so that it can be saved to a file, database, or transferred across a network. Deserialization is the reverse process of recreating the object from the byte stream. Implementing the marker interface 'java.io.Serializable' enables a class for serialization.

2. Code Example

```
import java.io.*;

public class SerializationDemo implements Serializable {
    private static final long serialVersionUID = 1L;
    String name = "Java Masterclass";

    public static void main(String[] args) {
        try (ObjectOutputStream oos = new ObjectOutputStream(new
        FileOutputStream("object.ser"))) {
            oos.writeObject(new SerializationDemo());
            System.out.println("Object serialized successfully.");
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 29. Serialization & Object Persistence?

- A) Option A for 29. Serialization & Object Persistence statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-

Topic 30: 30. Java Design Patterns (Singleton & Factory)

1. Theory

Design patterns are typical solutions to commonly occurring problems in software design. The Singleton pattern ensures a class has only one instance and provides a global point of access to it. The Factory design pattern creates objects without exposing the instantiation logic to the client, promoting loose coupling and adherence to SOLID principles.

2. Code Example

```
class Singleton {
    private static Singleton instance;
    private Singleton() {}

    public static synchronized Singleton getInstance() {
        if (instance == null) {
            instance = new Singleton();
        }
        return instance;
    }
}

public class DesignPatternDemo {
    public static void main(String[] args) {
        Singleton s1 = Singleton.getInstance();
        Singleton s2 = Singleton.getInstance();
        System.out.println("Same instance reference: " + (s1 == s2));
    }
}
```

3. Practice Questions (25 MCQs)

Q1. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q2. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q3. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q4. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q5. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q6. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q7. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q8. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q9. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q10. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q11. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q12. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q13. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q14. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q15. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q16. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q17. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q18. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q19. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q20. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q21. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q22. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q23. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q24. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

Q25. Which of the following best describes a key concept or correct syntax related to 30. Java Design Patterns (Singleton & Factory)?

- A) Option A for 30. Java Design Patterns (Singleton & Factory) statement
- B) Option B (Correct implementation behavior)
- C) Option C for alternative structure
- D) Option D for deprecated approach

4. Answer Key

Q.No	Correct Answer	Q.No	Correct Answer
Q1	B (Correct implementation behavior)	Q14	B (Correct implementation behavior)
Q2	B (Correct implementation behavior)	Q15	B (Correct implementation behavior)
Q3	B (Correct implementation behavior)	Q16	B (Correct implementation behavior)
Q4	B (Correct implementation behavior)	Q17	B (Correct implementation behavior)
Q5	B (Correct implementation behavior)	Q18	B (Correct implementation behavior)
Q6	B (Correct implementation behavior)	Q19	B (Correct implementation behavior)
Q7	B (Correct implementation behavior)	Q20	B (Correct implementation behavior)
Q8	B (Correct implementation behavior)	Q21	B (Correct implementation behavior)
Q9	B (Correct implementation behavior)	Q22	B (Correct implementation behavior)
Q10	B (Correct implementation behavior)	Q23	B (Correct implementation behavior)
Q11	B (Correct implementation behavior)	Q24	B (Correct implementation behavior)
Q12	B (Correct implementation behavior)	Q25	B (Correct implementation behavior)
Q13	B (Correct implementation behavior)		-